CSCI 690 [000769] Section 01, 3 Cr.
Research Methods in Computing
Spring 2026 SYLLABUS

INSTRUCTOR: Andrew A. Anda, Ph.D., Professor of Computer Science
Support: ECC223/CH366AE Telephone: (320)308-2044
E-MAIL: aanda@stcloudstate.edu (please use "CSCI604: " in the Subject field)
Instructor Schedule: https://aanda-stcloudstate-edu.github.io/WebSpace/Wkly-sched-w26.pdf
(and by appointment)
Student Support Hours Zoom: https://minnstate.zoom.us/j/91845917984
{I sometimes won't be available for a Student Support hour - this will usually be because I'm attending a meeting}

Resource Links:

- D2L Brightspace
-
- SCSU Student Resources for Online Learning
-
- SCSU Zoom Resources
-
- SCSU Student Information for Attending Classes Off-Campus
-
- SCSU: The Write Place
-
- SCSU Information Technology Services
-
- SCSU Medical Clinic
-
- SCSU Counseling and Psychological Services
-
- SCSU Library
-
- SCSU Student Code of Conduct
-

Additional SCSU student resources, curated and compiled by the Academic Affairs Support, are presented in:
Spring 2023 Student Instructional Resource and Support Guide

Content Delivery and Accessibility:
The delivery method is "On Campus".

CLASS TIME AND LOCATION:
Class: MWF: 11:00 - 11:50 in CH 9.

REQUIRED TEXTs: * Writing for Computer Science, 3rd ed., Justin Zobel, Springer, 2014.

* Measuring Computer Performance: A Practitioner's Guide, 1st ed., David J. Lilja, Cambridge University Press, 2005.

* Gift of Fire, A: Social, Legal, and Ethical Issues for Computing Technology, 5th ed., Sara Baase & Timothy Henry, Pearson, 2018.

SCSU COURSE CATALOG DESCRIPTION for CSCI690:

Formulation of a constrained computational research problem.
Measurement, display, and analysis of computational performance.
Tools, processes, and style conventions for writing and presenting a research paper.
Ethical considerations and obligations in the profession.
Preparation for writing and presenting a culminating project.

DESCRIPTION (*of the* one credit CSCI681 *course which this course evolved from*):

In practice, this course has evolved beyond the course catalog description.
This course is now taught as a technical writing and research methods course preparing the student for writing in the thesis style.
Towards this end, Students are expected to satisfactorily demonstrate competence in the execution of the following deliverables:

* use of a Reference Management System
* writing a research topic description as an extended abstract
* writing an annotated bibliography
* writing a research paper conforming to the SCSU graduate thesis style requirements
* writing a slide presentation and presenting it orally.

All of the writing deliverables must be coupled to the same research topic.

COURSE OUTLINE:

* Ethical considerations and obligations in the computing profession. 20%
* Measurement, display, and analysis of computational performance. 20%
* Academic writing style. 20%
* Reading, assessing, indexing, and annotating research literature. 5%
* Formulation and expression of a constrained computational research problem. 5%
* Editing and proofreading. 5%
* Scientific document preparation systems. 5%
* Other professional writing. 5%
* Research presentation. 10%
* Navigating the culminating project process. 5%

SLOs:

* Identify ethical considerations and obligations in the computing profession.
* Formulate a computational research problem.
* Measure, display, and analyze computational performance.

* Demonstrate an academic and technical writing and presentation process.

PREREQUISITES:
Graduate Standing in Computer Science

ACADEMIC HONESTY:
You are expected to do your own homework. If you copy someone else's work or allow someone else to copy your work, you are being academically dishonest and will be subject to severe disciplinary action which may include any or all of: no credit for the work in question, a failing grade for the course, notification to the university that you have violated your Code of Conduct. Use of recording devices during exams is prohibited. If you must quote or paraphrase another source, citation is essential, otherwise plagiarism has been committed. You are expected to be familiar with your rights and obligations as outlined in the "Code of Conduct"

[http://www.stcloudstate.edu/studenthandbook/documents/2013_14_codeofconduct.pdf] See also:
[http://www.stcloudstate.edu/studenthandbook/code/prohibited.asp]

[http://www.stcloudstate.edu/studenthandbook/code/rights_responsibilities.asp]

EXPLORATORY USE OF AI: (*Supportive but Limited*)
Supported tools such as Copilot and Adobe AI tools may be used in this course for limited academic tasks such as idea generation, outlining, or early-stage revisions.
Use of agentic AI tools integrated transparently into software products such as Grammarly and Overleaf are generally OK.
However, all submitted work must reflect your own understanding, voice, and academic reasoning.
AI tools should serve as a support, not a replacement, for your thinking.
Submitting work that is primarily AI-generated without acknowledgment is a breach of our Academic Integrity Policy.
If you use AI tools as part of your process, you must be transparent about how they were used (e.g., â€œUsed Copilot to draft bullet points; expanded and revised with personal analysisâ€).
When in doubt, ask for clarification before submitting your work.
Please consult the Academic Integrity Policy for more information and letâ€™s maintain an open dialogue about ethical and responsible technology use.

ATTENDANCE:
You are responsible for knowing what happens at each class meeting.

EXAMS:
There will be D2L Quizzes. No AI assistance is allowed when taking quizzes, but AI can be used to generate study materials.

GRADING:
Your grade will be determined by the contributions of exams, your assignments, term papers, and their oral (slide) presentations.

APPROXIMATE GRADE WEIGHTING:

Assignments	80%
Quizzes	20%

CAVEAT:
I reserve the right to amend the contents of this syllabus with notification.
****
**